

Solution Code - Module 3-Core Java - RecursiveFibonacci

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q13_RecursiveFibonacci.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q13_RecursiveFibonacci {
4 |     // Recursive method to find the nth Fibonacci number
5 |     // Sequence starts: fib(1)=0, fib(2)=1, fib(3)=1, fib(4)=2, ...
6 |     // Or fib(0)=0, fib(1)=1, fib(2)=1, ... We will use standard math: fib(1)=1, fib(2)=1, fib(3)=2, ...
7 |     public static int fibonacci(int n) {
8 |         if (n <= 0) {
9 |             return 0;
10 |        }
11 |        if (n == 1 || n == 2) {
12 |            return 1;
13 |        }
14 |        return fibonacci(n - 1) + fibonacci(n - 2);
15 |    }
16 |
17 |    public static void main(String[] args) {
18 |        Scanner scanner = new Scanner(System.in);
19 |        System.out.print("Enter a positive integer n: ");
20 |
21 |        if (scanner.hasNextInt()) {
22 |            int n = scanner.nextInt();
23 |            if (n <= 0) {
24 |                System.out.println("Error: Please enter a positive integer (greater than 0).");
25 |            } else {
26 |                // Avoid too large numbers to prevent stack overflow or slow execution
27 |                if (n > 40) {
28 |                    System.out.println("Warning: Numbers larger than 40 can take significant time to
calculate recursively.");
29 |                }
30 |                int result = fibonacci(n);
31 |                System.out.println("The " + n + "-th Fibonacci number is: " + result);
32 |            }
33 |        } else {
34 |            System.out.println("Error: Please enter a valid integer.");
35 |        }
36 |
37 |        scanner.close();
38 |    }
39 | }
```